CS 422/622 – Project Topic Selection

Project Topic Selection

Each team will identify an engaging, meaningful real-world problem that can be investigated using machine learning. The project should address an underexplored question or offer a new perspective on an existing problem; simply reproducing a standard Kaggle task or published tutorial is not sufficient.

Topic Requirements

- Start with the problem, not the algorithm. Clearly identify the real-world need, the people or organizations affected, and why the problem matters.
- Define a unified objective. State what the team intends to predict, classify, discover, or improve. All team members' work should contribute to this shared objective.
- Establish novelty. Explain how the project differs from common tutorials, existing solutions, or previous studies. Novelty may come from a new question, dataset, population, feature representation, evaluation criterion, or comparison of methods.
- Confirm data feasibility. Teams may use datasets from sources such as Kaggle or the UCI Machine Learning Repository, or another dataset relevant to their interests. The selected data must be accessible, sufficiently documented, and appropriate for the proposed objective.
- Consider responsible use. Identify potential concerns involving privacy, consent, bias, fairness, or harmful consequences, when applicable.
- Plan individual contributions. Each team member will implement and evaluate at least one machine-learning algorithm. These algorithms should be compared within the same experimental framework rather than treated as unrelated individual projects.

Topic Proposal

The proposal should briefly include:

1. The real-world problem and why it is important
2. The intended users or stakeholders
3. The machine-learning objective and target outcome
4. The proposed dataset and its source
5. How the topic differs from existing work or standard examples
6. One anticipated challenge or limitation
7. A preliminary plan for each team member's contribution

Additional Requirement for Graduate Students

Graduate students must conduct a brief review of relevant research literature. They should identify and study research papers closely related to the team's problem, dataset, or machine-learning methods.

Graduate students are expected to:

- Explain how the selected papers approach the problem.
- Compare the papers' datasets, features, algorithms, evaluation metrics, and experimental settings with the team's implementation.
- Reproduce or adapt at least one relevant method or baseline when reasonably feasible.
- Analyze similarities and differences between the published results and the team's results.
- Discuss possible reasons for differences, such as data preparation, parameter settings, sample size, evaluation procedures, or implementation choices.
- Identify at least one limitation or research opportunity suggested by the comparison.

The final report must cite the selected papers and clearly distinguish published methods and results from the team's own implementation and findings. A summary of the papers alone is not sufficient; the comparison should demonstrate critical analysis and connect the literature directly to the project.

Submission instructions:

Each team leader should submit the proposal, including the team's name and members by the given deadline.